INCH-POUND
A-A-59980
15 March 2017

# COMMERCIAL ITEM DESCRIPTION

# CALIBRATION SET, 1000 PSI PNEUMATIC DEADWEIGHT PRESSURE STANDARD

The General Services Administration has authorized the use of this commercial tem description for all federal agencies.

## 1. SCOPE

1.1 Scope. This Commercial Item Description (CID) specifies the Performance, Function, Design, and Environmental requirements for a Pneumatic Deadweight Pressure Standard (PDPS). The PDPS is primarily intended to calibrate pressure transfer standards and other pressure measuring devices using gas as the pressure medium. The item fills requirements currently addressed by an existing deadweight pressure standard. This PDPS will be the Air Force Primary Pressure Standard for gas pressure 1 pound per square inch (psi) to 1000 psi.

## 2. SALIENT CHARACTERISTICS

2.1 PDPS description. The PDPS is a deadweight piston gauge system covering a minimum range of 1 to 1000 psi in gauge and absolute modes of operation. The system includes a platform for mounting piston-cylinders onto which weights can be loaded, piston-cylinder assemblies, a mass set, means of reading necessary ambient and instrument conditions and calculating pressure, a means of adjusting pressure to slew between pressure points and float the piston and a vacuum pump to supply reference vacuum in absolute mode. Automation of operation shall be provided to reduce calibration time and operation influence on measurements.

2.1.1 The PDPS shall consist of an exchange package and a basic system. The exchange package shall include one complete set of piston-cylinders, one mass set, and reusable carrying/transit cases as required to facilitate shipment. The basic system shall include all other components, hardware, software, and accessories required to satisfy this CID.

2.2 Design and construction. The PDPS shall be designed and constructed in accordance with all applicable European Union (EU) requirements in order to have the “CE” marking affixed (see 2.2.2.2). The PDPS shall be designed and constructed so that no parts will work loosed in

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service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weather proof and designed to prevent the intrusion of water and sand into critical operating components.

2.2.1 Materials, protective coatings, and finish.

2.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes or climatic conditions or when exposed to heat shall not be used. Fasteners, handles, and fittings used in the assembly of the item shall also be primed and painted.

2.2.2 Markings.

2.2.2.1 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible, but shall not be less than ½ inch high. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of SAE-AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of SAE-AMS-STD-595. The center of gravity of the PDPS shall be stenciled on the unit within 1.0 inch of the calculated center of gravity.

2.2.3 Safety.

2.2.3.1 System safety. The design of the PDPS shall not contain any system safety mishap risk categories greater than medium as defined in MIL-STD-882.

2.2.3.2 Electrostatic discharge (ESD). The design of the PDPS shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmosphere due to sparking.

2.4 Performance requirements. The PDPS shall set and control absolute and gage pressure values for the purpose of calibrating other pressure measuring devices. The PDPS shall operate in manual and semi-automated modes.

2.5 Pressure range and uncertainties.

2.5.1 Pressure range. The PDPS shall set and control absolute and gauge pressure in the range of at least 1 to 1000 psi. Multiple piston-cylinders elements and one mass set may be used to cover the range. The ranges of pressure covered by the piston-cylinders shall overlap such that any pressure range with a range of 15 psi or greater can be calibrated in 10% increments with a single piston-cylinder.



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2.5.2 Measurement uncertainty. The measurement uncertainty of the PDPS in absolute and gauge measurement modes shall be equal to or better than +/- (0.00003 psi + 15 parts per million(ppm) of set pressure) from 1 to 50 psi, +/- (0.00008 psi + 20 ppm of set pressure) from 50 to 250 psi, +/-(0.0005 psi + 25 ppm of set pressure) from 250 to 1000 psi. The PDPS shall include the secondary measurement capabilities of ambient and instrument conditions necessary to support the required uncertainty level.

2.5.2.1 The measurement uncertainty provided shall be supported by an uncertainty analysis following the ISO Guide to Uncertainty in Measurement (GUM) and NIST TN1297. The analysis shall include, at a minimum, consideration of piston-cylinder effective area, piston-cylinder temperature, piston-cylinder verticality, resolution, linearity, mass, local gravity, air buoyancy (gauge mode), reference vacuum measurement (absolute mode), fluid head and ambient conditions in the calibration laboratory. The uncertainty analysis shall be provided with proposal.

2.5.2.2 The PDPS measurement uncertainty shall be based on a 95% confidence interval (probability) that the next PDPS measurement will be within the measurement uncertainty limit of the true pressure.

2.5.3 Stability over time. The PDPS shall remain within the measurement uncertainty required in 2.5.2 for a minimum of 30 months with recalibration. The uncertainty analysis in 2.5.2.1 shall include an analysis of stability. Re-zeroing of vacuum gauges or measurements devices is acceptable.

2.5.4 Maintenance of measurement uncertainty. The recalibration of PDPS will be supported by the Air Force Primary Standards Laboratory (AFPSL). A description of the hardware and methodology required to maintain the measurement uncertainty provided in 2.5.2 shall be provided. The presentation shall include a discussion of means for maintaining traceability and time required to perform the recalibration function.

2.6 Piston-cylinders.

2.6.1. Material. Piston-cylinders shall be made of a material whose Young's Modulus and Thermal Expansion Coefficient are adequate to assure the maintenance of the measurement uncertainty and stability requirement of 2.5.1 and 2.5.2.

2.6.2 Fall rate. Once floated and stabilized, the piston shall continuously control the set pressure without readjustment for a minimum of 10 minutes from 1 to 50 psi, 5 minutes from 50 to 250 (psi) and 2 minutes from 250 to 1000 psi.

2.6.3 Piston-cylinder cleaning. When operated with a clean, dry (instrument grade) Nitrogen of Air supply, all piston–cylinders shall operate continuously within the measurement uncertainty specifications for a minimum of 8 hours without requiring cleaning.

2.6.4 Reference vacuum. To minimize the uncertainty associated with the measurements of reference vacuum, all piston-cylinders shall operate in absolute mode within the performance



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limits of section 2.4 when exposed to a reference vacuum of 0.0005 psi or lower. Control or adjustment shall not be required other than waiting for pull down to reach the limits of 2.9

2.7 Mass set.

2.7.1 Mass material. Masses shall be machined from non-magnetic stainless steel individual masses shall be solid without adjusting cavities or parts.

2.7.2 Mass set makeup. The mass set shall include sufficient mass to cover the required pressure range. Its composition shall allow pressure value within the range to be set. The resolution shall be sufficiently high to allow the exact pressure to be set within 10 ppm or less of any nominal pressure value in the range.

2.7.3 Mass values. Individual masses shall be in nominal multiples and sub multiples of kilograms. The serial number of the mass set and individual mass number shall be engraved on the masses.

2.8 Pressure control.

2.8.1 Manual pressure control range. The PDPS shall set and control pressure over the range of 1 to 1000 psi absolute and gauge pressure. The PDPS pressure control component shall be capable of slewing pressure between pressure points, floating the piston at the set pressure and re-floating as needed.

2.8.2 Pressure control automation. Automated pressure control to slew the pressure and float the piston at a requested set pressure shall be implemented. Automated pressure control shall cover the complete pressure range. This shall be proposed as a complete system.

2.8.3 Direct pressure control. Direct operator control to slew pressure and adjust the position of the floating piston is required. Direct pressure control shall include a directly accessible means of rapidly venting the measured pressure to atmosphere at any time.

2.8.4 Stability of set pressure. The PDPS shall set absolute pressure values over the complete range with a stability of +/- 5 ppm of set pressure or 0.00001 psi, whichever is greater, over the range of 1 to 1000 psi.

2.8.5 Pressure slew rate. The maximum slew rate of the pressure controlling device shall be such that a 1 to 1000 psi excursion, for example to exercise the device under test, can be completed with a test volume of 250 cubic centimeters (cc) in less than 20 seconds. The operator shall be able to adjust the pressure without detriment to the long or short term performance of the piston-cylinder.

2.8.6 Pressure setting time. The time required to complete a 10% calibration increment within the range, from a stable floating piston at one pressure to a stable floating piston at another pressure, with a test volume of 250 cc or less, shall be less than 90 seconds for a 1000 psi range, less than 60 seconds for a 250 psi range and less than 30 seconds for a 50 psi range. The rate at



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which pressure is changed when slewing from calibration point to calibration point shall not affect the behavior of the piston.

2.8.7 Test volume. The PDPS shall meet all of the pressure generation and control requirements of 3.5 when connected to leak free test volumes from 0 to 250 cc.

2.8.8 Pressurized gas. The PDPS shall be able to operate within all performance limits using clean, dry, instrument grade nitrogen or air as the pressurized fluid.

2.8.9 Control vacuum supply. The pressure control component shall be able to set pressure from 1 to 1000 psi in absolute mode with a vacuum supply of 0.1 pound per square inch absolute [psia].

2.9 Reference vacuum pump. A reference vacuum pump package shall be included in the PDPS. If the pump is oil sealed it shall include an oil vapor trap. The package shall also include the necessary hardware to connect the pump to the piston gauge vacuum reference port and to isolate the pump from the vacuum port when releasing vacuum from the bell jar.

2.9.1 Reference vacuum level and pull-down time. The vacuum achieved shall be of less than 20 millitor (mTorr) and 1 to 5 psia and less than 30 mTorr above 50 psia. The time required to reach the reference vacuum level shall be less than 4 minutes for an initial absolute pressure point and less than 2 minutes for subsequent points.

2.10 Design requirements. The PDPS is composed of a deadweight piston gauge platform, piston-cylinder assemblies, a mass set, a pressure control component, readout and means or reading necessary ambient and instrument conditions and calculating pressure. Automation of operation to reduce calibration time and operator influence on measurements shall be implemented. Operator ergonomics shall be maximized.

2.10.1 Size and weight. The overall PDPS bench top footprint, including storage of masses when not loaded, shall not exceed 8 square feet (2 ft x 4 ft). The total weight of the PDPS shall not exceed 100 kilograms, not including the vacuum pump.

2.11 Electrical power and connections. The PDPS components shall operate from a source of 103.5 to 129 volts at 50 to 60 H, +/- 5% single phase input power. A detachable power cord shall be provided with the system.

2.12 Warm up. The PDPS shall meet all requirements contained in this purchased description after a warm up time not to exceed 30 minutes. This does not include the time required for the piston-cylinder to be acclimated after cleaning.

2.13 User interface. All PDPS regular operating modes and functions shall be accessible using this interface. The PDPS shall interface with either a laptop computer via RS232 or a piston gauge monitor.



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2.13.1 User interface display. The user interface shall provide real time indications of:

a. Current measure pressure, then have a “ready light”, when the piston is floating within the measurement uncertainty required by 3.5.2.

b. True mass load with resolution of 0.01 grams.

c. Piston position with resolution 0.01millimeter.

d. Piston rotational rate with resolution of 1 revolution per minute.

e. Piston fall rage with resolution of 0.01mm/min.

f. Reference vacuum in absolute mode with resolution of 0.1 mTorr.

g. Piston rotation rate deceleration with resolution of 1 revolution per minute.

h. Piston temperature with resolution of 0.01 degrees Celsius.

i. Ambient temperature with resolution of 0.1 degrees Celsius.

j. Ambient relative humidity with resolution of 1%.

k. Ambient pressure with resolution of 0.001 psi.

2.13.2 User interface modes and functions. The user interface shall support, at a minimum, the following modes and functions:

a. Pressure entry mode and mass entry modes: Pressure entry allows the operator to enter a pressure and instructs the operator which mass to load. Mass entry allows the operator to enter a mass load and displays the resulting measured pressure.

b. Unit of measure selection.

c. Measurement mode selection (gauge, absolute).

2.14 Pressure unit of measure. The PDPS shall have, as a minimum, the following engineering units available on screen for operator selection: Psi, in mercury, inches of water at various temperatures, Torr (Torr), Pascal (Pa), Kilopascal (kPa), Barometer (bar).

2.15 Operating modes. The PDPS shall have absolute and gauge pressure modes selectable via the User Interface. Changing from one operating mode to the other shall be able to be completed in less than 10 minutes.

2.16 Piston rotation. A means of maintaining the piston within the prescribed rate of rotation and avoiding making measurements outside of rotation rate limits shall be provided.



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2.17 Calibration.

2.17.1 Configuration. The PDPS piston-cylinder and mass sets shall be easily removed from the PDPS system for transportation to a calibration facility. If the deadweight piston gauge platform is needed for the calibration, it shall also be easily removed and shippable. Secondary measurement devices such as the reference vacuum gauge and piston-cylinder temperature sensor shall be easily removed from the system.

2.17.2 Carry cases. Reusable transit cases shall be provided for PDPS components that require mass or cross float calibration or other offsite calibration, including the deadweight piston gauge platform if necessary. The cases shall provide protection against shock and vibration during shipment via Government or commercial carrier.

2.17.3 Calibration data. The PDPS shall provide a means to change any calibration or standardization data stored in memory. The PDPS shall provide a means of data protection such that during normal operation this data cannot be corrupted.

2.18 Environment. The PDPS shall operate within the requirements of section 2 in the environment of a typical, high precision, physical metrology laboratory.

2.18.1 Operating temperature. The PDPS shall meet the requirements of section 2 over the operating temperature range of 15 to 25 Degree Celsius.

2.18.2 Operating humidity. The PDPS shall meet the requirements of section 2 over the operating humidity range of 15 % to 70%.

2.19 System computer. Computer supplied with the system shall be notebook type with the latest Windows 7 operating system. The system computer shall include a DVD/CD drive for archiving.

2.19.1 Software requirements. The software supplied shall include an opening “splash screen” that clearly identifies the software and version. The software identification and revision shall be clearly marked on the install disk(s) that shall be provided with the system. It shall provide a single use license for use at a single location.

2.19.2 Software maintenance. The contractor shall notify the Air Force of upgrades to the software as they become available.

2.19.3 Cross-float. A recommended cross-float calibration procedure shall be provided with the first two PDPS systems.

2.19.4 Cross-float software. A program for cross-floating shall be provided that will display on a laptop computer the necessary parameters to perform a successful cross-float. The capability to monitor on-board sensors and display sink rates of the two pistons shall be provided. RS232 interface is the preferred method for interfacing with the dead weight gauge. The software shall



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have the capability of determining an Ao of the piston once a viable cross-float is completed. (Ao is the area of the piston at zero or non-distorted pressure.)

2.20 Workmanship. The PDPS, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs, and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

## 3. REGULATORY REQUIREMENTS

3.1 Recovered, or environmentally preferable, or biobased materials. The offeror/contractor is encouraged to use recovered materials or biobased materials to the maximum extent practicable, in accordance with the Federal Acquisition Regulation (FAR), Part 23.403. However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

## 4. PRODUCT CONFORMANCE

4.0 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. PDPS inspection (see 4.2)

b. Operational inspection (see 4.3)

c. Configuration inspection (see 4.4)

Salient characteristics shall be verified in accordance with Table I.



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TABLE I Salient characteristics verification matrix.

| Section 2 Salient Characteristics | Verification Method | Section 4 Product Conformance Provisions |
|---|---|---|
| 2.1 PDPS description. | N/A | |
| 2.2 Design and construction. | Examination | 4.6.1 Examination of product. |
| 2.2.1 Materials, protective coatings, and finish. | N/A | |
| 2.2.1.2 Protective coatings. | Examination | 4.6.1 Examination of product. |
| 2.2.2 Markings. | N/A | |
| 2.2.2.1 Markings. | Examination | 4.6.1 Examination of product. |
| 2.2.3 Safety. | N/A | |
| 2.2.3.1 System safety. | Analysis | 4.6.2 System safety hazard analysis |
| 2.2.3.2 Electrostatic discharge (ESD). | Analysis | 4.6.3 Electrostatic discharge analysis. |
| 2.4 Performance requirements. | Test | 4.6.4 Performance requirements test. |
| 2.5 Pressure range and uncertainties. | N/A | |
| 2.5.1 Pressure range. | Test | 4.6.5 Pressure range test. |
| 2.5.2 Measurement uncertainty. | Analysis | 4.6.6 Measurement uncertainty analysis. |
| 2.5.3 Stability over time. | Analysis | 4.6.6 Measurement uncertainty analysis. |
| 2.5.4 Maintenance of measurement uncertainty. | Analysis | 4.6.6 Measurement uncertainty analysis. |
| 2.6 Piston-cylinders. | N/A | |
| 2.6.1 Material. | Analysis | 4.6.7 Material analysis. |
| 2.6.2 Fall rate. | Test | 4.6.8 Fall rate test. |
| 2.6.3 Piston-cylinder cleaning. | Test | 4.6.9 Piston-cylinder cleaning test. |
| 2.6.4 Reference vacuum. | Test | 4.6.10 Reference vacuum test. |
| 2.7 Mass set. | N/A | |
| 2.7.1 Mass material. | Examination | 4.6.1 Examination of product. |
| 2.7.2 Mass set makeup. | Demonstration | 4.6.11 Mass makeup demonstration. |
| 2.7.3 Mass values. | Examination | 4.6.1 Examination of product. |
| 2.8 Pressure control. | N/A | |
| 2.8.1 Manual pressure control range. | Test | 4.6.12 Manual pressure control range test. |
| 2.8.2 Pressure control automation. | Demonstration | 4.6.13 Pressure control demonstration. |



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TABLE I Salient characteristics verification matrix – Continued.

| Section 2 Salient Characteristics | Verification Method | Section 4 Product Conformance Provisions |
|---|---|---|
| 2.8.3 Direct pressure control. | Test | 4.6.14 Direct pressure control test. |
| 2.8.4 Stability of set pressure. | Test | 4.6.15 Stability of set pressure test. |
| 2.8.5 Pressure slew rate. | Test | 4.6.16 Pressure slew rate test. |
| 2.8.6 Pressure setting time. | Test | 4.6.17 Pressure setting time test. |
| 2.8.7 Test volume. | Test | 4.6.18 Test volume test. |
| 2.8.8 Pressurized gas. | Test | 4.6.19. Pressurized gas test. |
| 2.8.9 Control vacuum supply. | Test | 4.6.20 Control vacuum supply test. |
| 2.9 Reference vacuum pump. | Examination | 4.6.1 Examination of product. |
| 2.9.1 Reference vacuum level and pull-down time. | Test | 4.6.21 Reference vacuum level and pull-down time test. |
| 2.10 Design requirements. | Examination | 4.6.1 Examination of product. |
| 2.10.1 Size and weight. | Examination | 4.6.1 Examination of product. |
| 2.11 Electrical power and connections. | Test | 4.6.22 Electrical power and connections test. |
| 2.12 Warm up. | Test | 4.6.23 Warm up test. |
| 2.13 User interface. | Test | 4.6.24 User interface test. |
| 2.13.1 User interface display. | Test | 4.6.25 User interface display test. |
| 2.13.2 User interface modes and functions. | Test | 4.6.26 User interface modes and functions test. |
| 2.14 Pressure unit of measure. | Examination | 4.6.1 Examination of product. |
| 2.15 Operating modes. | Demonstration | 4.6.27 Operating modes demonstration. |
| 2.16 Piston rotation. | Examination | 4.6.1 Examination of product. |
| 2.17 Calibration. | N/A | |
| 2.17.1 Configuration. | Examination | 4.6.1 Examination of product. |
| 2.17.2 Carry cases. | Examination | 4.6.1 Examination of product. |
| 2.17.3 Calibration data. | Test | 4.6.28 Calibration data test. |
| 3.18 Environment. | Examination | 4.6.1 Examination of product. |
| 2.18.1 Operating temperature. | Test | 4.6.29.1 Operation temperature test. |



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TABLE I Salient characteristics verification matrix – Continued.

| Section 2 Salient Characteristics | Verification Method | Section 4 Product Conformance Provisions |
|---|---|---|
| 2.18.2 Operating humidity. | Test | 4.6.29.2 Operation humidity test. |
| 2.19 System computer. | Examination | 4.6.1 Examination of product. |
| 2.19.1 Software requirements. | Demonstration | 4.6.30 Software requirements demonstration. |
| 2.19.2 Software maintenance. | N /A | |
| 2.19.3 Cross-float. | Examination | 4.6.1 Examination of product. |
| 2.19.4 Cross-float software. | Demonstration | 4.6.31 Cross-float software demonstration. |
| 2.20 Workmanship. | Examination | 4.6.1 Examination of product. |
| 3.1 Recovered, or environmentally preferable, or biobased materials. | Examination | 4.6.1 Examination of product. |

4.2 PDPS inspection. The PDPS shall be subjected to the analyses, demonstrations, examinations, and tests described in 4.6.1 through 4.6.31. The contractor shall provide or arrange for all test equipment and facilities. Unless otherwise approved by the procuring activity, all PDPS items shall be in the same configuration at all times and configuration changes shall not be made during the PDPS inspection.

4.3 Operational inspection. All operational inspection shall be satisfactorily completed and approved prior to final PDPS inspection.

4.4 Configuration inspection. Each production PDPS shall be subjected to the examination described in 4.6.1.

4.5 Inspection requirements.

4.5.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals in ensure laboratory accuracy.

4.5.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.



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d. Ambient temperature.

e. Ambient humidity

4.5.3 Test rejection criteria. Throughout all tests specified herein, the PDPS shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein, or in the contractor's technical proposal.

b. Structural failure of any component, including permanent deformation, or evidence or impending failure.

c. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer's (OEM's) specifications or tolerances shall be utilize for making determination.

d. Evidence or corrosion or deterioration.

e. Misalignment of components

f. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

4.5.4 Failure reporting, analysis, and corrective action system (FRACAS). A closed loop system shall be used to collect data, analyze, and record timely corrective action for all potential failures that occur during the PDPS and operational test. The contractor's existing FRACAS shall be utilized with the minimum changes necessary to conform to this specification. The system shall cover all test samples, interfaces between test samples, test instrumentation, test facilities, test procedures, test personnel, and the handling and operating instructions. The contractor shall establish and maintain a FRACAS database; all information shall be entered into the database. Authorized procuring activity personnel shall have read-only access to the FRACAS database.

4.5.4.1 Problem and failure action. At the occurrence of a problem or potential failure that affects satisfactory operation of a test sample, entries shall be made in the appropriate data logs and the failed test sample shall be removed from test, with minimum interruption to the other test samples continuing on test.

4.5.4.1.1 Problem and failure reporting. A failure report shall be initiated at the occurrence of each problem or potential failure of the contractor hardware or software or government furnished equipment (GFE). The report shall contain the information required to permit determination of the origin and correction of the failure. The contractor's existing failure report form may be used with minimum changes necessary to conform to the requirements of this specification. The failure report shall include the information specified in "a through c":



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a. Descriptions of failure symptoms, conditions surrounding the failure, failed hardware identification, and operating time (or cycles) at the time of failure.

b. Information of each independent and dependent failure and extent of confirmation of the failure symptoms, the identification of failure modes, and a description of all repair actions taken to return the test sample to operational readiness.

c. Information describing the results of the investigation, the analysis of all part failures, an analysis of the system design, and the corrective action taken to prevent failure recurrence. If no corrective action is taken, the rational for this decision shall be recorded.

4.5 4.1.2 Identification and control of failed items. A failure tag shall be affixed to the failed part immediately upon the detection of any failure or suspected failure. The failure tag shall provide space for the failure report serial number and for other pertinent entries from the test sample failure record. All failed parts shall be marked conspicuously or tagged and controlled to ensure disposed in accordance with contract requirements. Failed parts shall not be handled in any manner which may obliterate facts which might be pertinent to the analysis. Failed parts shall be stored pending disposition by the authorized approval agency of the failure analysis.

4.5.4.1.3 Problem and failure investigations. An investigation and analysis of each reported failure shall be performed. Investigation and analysis shall be conducted to the level of hardware of software necessary to identify causes, mechanisms, and potential effects of the failure. Any applicable method (for example, test, microscopic analysis, applications study, dissection, X-ray analysis, spectrographic analysis) of investigation and analysis which may be needed to determine failure cause shall be used. When the removed part is not defective or the cause of failure is external to the part, the analysis shall be extended to include the circuit, higher hardware assembly, test procedures, and subsystem if necessary. Investigation and analysis of GFE failures shall be limited to verifying that the GFE failure was not the result of the contractor's hardware, software, or procedures. This determination shall be documented for notification of the procuring activity.

4.5.4.1.4 Failure verification. Reported failures shall be verified as actual failures or an acceptable explanation provided to the procuring activity for lack of failure verification. Failure verification is determined either by repeating the failure mode of the reported part or by physical or electrical evidence of failure (for example, leakage residue or damaged hardware). Lack of failure verification, by itself, is not sufficient rationale to conclude the absence of a failure.

4.5.4.1.5 Corrective action. When the cause of failure has been determined, a corrective action shall be developed to eliminate or reduce the recurrence of the failure. Repairs shall be made in accordance with normal field operating procedures and manuals. The procuring activity shall review the corrective actions at the scheduled test status review prior to implementation. In all cases the failure analysis and the resulting corrective actions shall be documented.

4.5.4.1.6 Problem and failure tracking and closeout. The closed loop failure reporting system shall include provisions for tracking problems, failures, analysis, and corrective actions. Status or



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corrective actions for all problem and failures shall be reviewed at scheduled test status reviews. Problem and failure closeout shall be reviewed to assure their adequacy.

4.6 Test methods. PDPS testing shall be conducted by the contractor. Tests shall be performed to assure the PDPS meets or exceeds all of the requirements of this CID. The PDPS will also be inspected and accepted at the Air Force Primary Standards Laboratory (AFPSL). The item will be checked with test equipment and parameters equal to or better than those required assuring specified performance requirements are met. Test equipment has first echelon certification directly traceable to the National Institute of Standards and Technology (NIST) shall be required.

4.6.1 Examination of product. Each PDPS shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or test listed in 4.6. A contractor generated, Government–approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to material, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each PDPS function shall be verified. Certification and analyses shall be provided in accordance with Table II. Each production PDPS shall be inspected to a Government approved reduce version of the checklist.

TABLE II. Certifications and analyses.

| Paragraph | Required Certifications and Analyses |
|---|---|
| 3.3 Design and construction. | Contractor documentation that the PDPS is in accordance with all applicable EU requirements in order to have the “CE” marking affixed. |
| 3.3.3.1 System safety. | Contractor system safety hazard analysis (see 4.6.2) |
| 3.3.3.2 Electrostatic discharge. | Contractor analysis of the electrostatic discharge requirement (see 4.6.3) |
| 3.5.2 Measurement uncertainty.<br>3.5.3 Stability over time.<br>3.5.4 Maintenance of measurement uncertainty. | Contractor analysis of the measurement uncertainty (see 4.6.6) |
| 3.6.1 Material. | Contractor analysis of the materials used (see 4.6.7) |

4.6.2 System safety hazard analysis. A system safety hazard analysis of the PDPS shall be conducted in accordance with MIL-STD-882 to demonstrate compliance with the mishap risk requirement of 2.2.3.1.

4.6.3 Electrostatic discharge analysis. An engineering analysis shall be performed to demonstrate compliance with the electrostatic discharge requirement of 2.2.3.2.



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4.6.4 Performance requirements test. The PDPS shall be tested to demonstrate compliance with 2.4.

4.6.5 Pressure range test. The PDPS shall be tested to demonstrate compliance with 2.5.1.

4.6.6 Measurement uncertainty analysis. An engineering analysis shall be performed to demonstrate compliance with 2.5.2, 2.5.3, and 2.5.4.

4.6.7 Material analysis. An engineering analysis shall be performed to demonstrate compliance with 2.6 1.

4.6.8 Fall rate test. The PDPS shall be tested to demonstrate compliance with 2.6.2.

4.6.9 Piston-cylinder cleaning test. The PDPS shall be tested to demonstrate compliance with 2.6.3.

4.6.10 Reference vacuum test. The PDPS shall be tested to demonstrate compliance with 2.6.4.

4.6.11 Mass makeup demonstration. The PDPS shall be used to demonstrate compliance with 2.7.2.

4.6.12 Manual pressure control range test. The PDPS shall be tested to demonstrate compliance with 2.8.1.

4.6.13 Pressure control demonstration. The PDPS shall be tested to demonstrate compliance with 2.8.2.

4.6.14 Direct pressure control test. The PDPS shall be tested to demonstrate compliance with 2.8.3.

4.6.15 Stability of set pressure test. The PDPS shall be tested to demonstrate compliance with 2.8.4

4.6.16 Pressure slew rate test. The PDPS shall be tested to demonstrate compliance with 2.8.5.

4.6.17 Pressure setting time test. The PDPS shall be tested to demonstrate compliance with 2.8.6.

4.6.18 Test volume test. The PDPS shall be tested to demonstrate compliance with 2.8.7.

4.6.19 Pressurized gas test. The PDPS shall be tested to demonstrate compliance with 2.8.8.

4.6.20 Control vacuum supply. The PDPS shall be tested to demonstrate compliance with 2.8.9.

4.6.21 Reference vacuum level and pull-down time test. The PDPS shall be tested to demonstrate compliance with 2.9.1.



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4.6.22 Electrical power and connections test. The PDPS shall be tested to demonstrate compliance with 2.11.

4.6.23 Warm up test. The PDPS shall be tested to demonstrate compliance with 2.12.

4.6.24 User interface test. The PDPS shall be tested to demonstrate compliance with 2.13.

4.6.25 User interface display test. The PDPS shall be tested to demonstrate compliance with 2.13.1.

4.6.26 User interface modes and functions test. The PDPS shall be tested to demonstrate compliance with 2.13.2

4.6.27 Operating modes demonstration. The PDPS shall be tested to demonstrate compliance with 2.15.

4.6.28 Calibration data test. The PDPS shall be tested to demonstrate compliance with 2.17.3

4.6.29 Environment tests.

4.6.29.1 Operating temperature test. The PDPS shall be tested to demonstrate compliance with 2.18.1.

4.6.29.2 Operating humidity test. The PDPS shall be tested to demonstrate compliance with 2.18.2.

4.6.30 Software requirements demonstration. The PDPS shall be tested to demonstrate compliance with 2.19.1.

4.6.31 Cross-float software demonstration. The PDPS shall be tested to demonstrate compliance with 2.19.4.

## 5. **PACKAGING**

5.1 For acquisition purposes, the packaging requirements shall be as specified in the contract or order.

## 6. **NOTES**

6.1 Source of documents.

6.1.1 Government documents. Military Standards and Federal documents referenced herein are available online at http://quicksearch.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094



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6.1.2 SAE-AMS-STD-595. Colors used in government procurement are available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096 or online at http://standards.sae.org/amsstd595/

6.1.3 The Federal Acquisition Regulation (FAR). The FAR may be obtained from the Superintendent of documents, P.O. Box 371954, Pittsburgh PA, 15250-7954 or online at https://www.acquisition.gov/far .

6.1.4 National Institute of Standards and Technology (NSIT). NSIT documents can be obtained online at https://www.nist.gov/pml/weights-and-measures or from NSIT, 100 Bureau Drive, Gaithersburg, MD 20899

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, revision and date of this CID.

b. Packaging requirements (see 5.1).

6.3 National Stock Number (NSN). The following is a list of NSN's assigned that correspond to this CID. The list may not be indicative of all possible NSN's associated with this CID.

| NSN |
|---|
| 6695-01-616-5905 |
| 6695-01-551-5125 |

6.4 Key words.

Exchange package
Gas pressure
Gas Transfer
Piston gauge system
Pressure transfer



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**Concluding material:**

MILITARY INTEREST

**Custodians**:
Air Force - 84

**Preparing Activity**:
Air Force - 84

**Reviewer**:
Army - MI
Air Force - 20

**Agent**:
Air Force – 99
(Project 6695-2017-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .



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